

How Badly Did IT Screw You: The Full List

MEP Automation tool impact audit following the IT-deployed pyRevit 6.5.3 update — verified root causes and remediation plan

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Executive Summary

The recent IT-deployed pyRevit update (to version 6.5.3) introduced two breaking changes that disabled several in-house MEP automation tools. The failures were silent — tools presented blank or unresponsive windows with no error message, because the update also broke the error-reporting window itself.

Status: The flagship circuiting tool (SuperCircuit V5) has been fully diagnosed, repaired, and verified in production (86 circuits created on the PF North Asheville model). A repo-wide audit identified **7 additional tools confirmed or near-certain broken, 5 tools requiring a quick verification**, and one tool family (MEPRFP 2.0 panel) requiring a smoke test. The repair pattern is proven and mechanical; estimated remaining effort is **approximately one half-day** including per-tool verification.

Root Causes (verified)

- **Logger API removal.** pyRevit 6.x replaced its logger object; the standard `setLevel` call our older scripts use now throws an error that kills the script before any window opens. Verified on SuperCircuit V5; 4 affected files found and fixed.
- **Script-scope teardown ("scope wipe").** pyRevit 6.x now erases a script's in-memory state the moment the script finishes. Any tool with a *modeless* window (one that stays open while the user works in Revit) has its buttons and dropdowns go dead — clicks silently do nothing. This occurs *even when the tool declares a persistent engine*, which previously prevented it. Verified by instrumented testing; a workaround (state snapshot/restore) was developed and proven on SuperCircuit V5.
- **Secondary issue:** the pyRevit output/error window rendered fully blank for part of the session, masking all error messages and significantly extending diagnosis time. It recovered after a pyRevit reload and should be monitored.

Impact by Tool

Tool	Location	Status	Issue / Action
SuperCircuit V5	MEP Automation > Electrical	FIXED & VERIFIED	Both root causes; fully repaired, verified end-to-end (86 circuits created).
SuperCircuit V4	MEP Automation > Electrical	FIXED	Logger crash only (modal tool); repaired.

AI SuperCircuit V5	MEP Automation › Electrical	PARTIALLY FIXED	Logger crash repaired; still needs the modeless-window workaround (Run button currently dead).
Place Single Profile	MEP Automation	BROKEN	Modeless panel + external events; controls dead. Needs standard repair.
Check Synced Relationships	MEP Automation	BROKEN	Modeless conflicts window + external events; controls dead.
QAQC	MEP Automation › Audit Tools	BROKEN	Modeless window + external events; controls dead.
Manage Space Profiles	MEP Automation › Space Operations	BROKEN	Modeless window with hide/re-show cycle; controls dead.
Alerts Manager	CED ElecTools	BROKEN	Modeless singleton window + event gateway; controls dead.
Element 3D Rotation	CED MechTools › Piping	BROKEN	Modeless options window; already declared persistent engine, which no longer protects it.
Circuit Manager	CED ElecTools	VERIFY	Opens modal (should survive), but drives external events whose timing needs a one-click check.
Batch Swap Circuits	CED ElecTools	VERIFY	Modal-first window; low risk, quick check recommended.
Add Spares and Spaces	CED ElecTools	VERIFY	Modal-first window; low risk, quick check recommended.
Family Database	Family Operations	VERIFY	Hide/re-show window pattern; no external events; low risk.
Select / Get Hosted Tags	Tags panel	VERIFY	Relies on persistent state between clicks, which the scope wipe erases; pairing may silently reset.
MEPRFP Automation 2.0 panel (Place from CAD, Place Space Elements, SuperCircuit V5 2.0, Sync Audit)	AE pyTools	SMOKE TEST	Runs on the CPython engine, whose callback mechanics differ; a 5-minute click-through of one modeless tool determines whether the flagship panel is affected.
All modal/dialog-only tools (SuperCircuit V3, Audit Circuits, Circuit By Space, pickers, H-E-B / WM toolsets, transaction-only buttons)	All extensions	NOT AFFECTED	No modeless windows or removed APIs in use; repo-wide scan found no other victims.

Remediation Plan & Effort

- **Proven repair recipe** (validated on SuperCircuit V5): declare engine persistence in both the script and bundle configuration, and add a small state snapshot/restore guard (~20 lines) invoked at each UI event entry. Purely additive; backward-compatible with older pyRevit versions.
- **Remaining work:** apply the recipe to the 7 confirmed/near-certain broken tools and run one-click verifications on the 5 "verify" items and the 2.0 panel. **Estimate: ~half a day.**
- **Recommendation:** pin the pyRevit version in future IT update rounds, or route pyRevit updates through a one-model regression pass of the automation toolset before firm-wide deployment.

Methodology: root causes were isolated via instrumented breadcrumb tracing inside the Revit session (bypassing the broken error window), syntax-verified against pyRevit's bundled IronPython 2.7.12 engine, and confirmed by an end-to-end production run. The tool inventory was produced by a repo-wide scan for the two failure signatures (removed logger API; modeless windows / external event handlers without the new workaround) across all CED extensions on the MEP-AUTOMATION-TOOLS branch.